

Autodesk® Scaleform®

Scaleform 4.4

Scaleform 4.4

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Autodesk® Scaleform® 4.4

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1

Scaleform 4.0

(Advance)

Scaleform SDK

Scaleform 4.0 API

Scaleform

2

API Scaleform3.x Scaleform4

1. Renderer2D
2. GfxLoader
- 3.
4. MovieDisplayHandle
Renderer2D::Display
5. Gfx::Movie
6. API
Render::HAL
GRenderer Scaleform 3.X

2.1

Scaleform Render::Renderer2D Render::HAL Renderer2D
Display Render::HAL Renderer2D
" " Render::D3D9::HAL
Render::PS3::HAL

```
D3DPRESENT_PARAMETERS    PresentParams;  
IDirect3DDevice9*          pDevice = ...;  
ThreadId                  renderThreadId = Scaleform::GetCurrentThreadId();  
Render::ThreadCommandQueue *commandQueue = ...;  
Ptr<Render::D3D9::HAL>    pHal;  
Ptr<Render::Renderer2D>    pRenderer;  
  
pHal = *SF_NEW Render::D3D9::HAL(commandQueue);  
if (!pHal-> InitHAL(Render::D3D9::HALInitParams(pDevice, PresentParams, 0,
```

```

renderThreadId)))
{
    return false;
}
pRenderer = *SF_NEW Render::Renderer2D(pHal);

```

HALInitParams class

pDevice PresentParams

renderThreadId

ThreadCommandQueue

Renderer2D HAL

2.2

Scaleform4

Renderer2D

Scaleform 3.x

Loader

InitHAL

Render::HAL

ShutdownHAL

Render::HAL

HAL

```

MeshCacheParams mcp;
mcp.MemReserve      = 1024 * 1024 * 3;
mcp.MemLimit        = 1024 * 1024 * 12;
mcp.MemGranularity  = 1024 * 1024 * 3;
mcp.LRUTailSize     = 1024 * 1024 * 4;
mcp.StagingBufferSize = 1024 * 1024 * 2;

pRenderer->GetMeshCacheConfig()->SetParams(mcp);

```

MemReserve

MemLimit

MeshCache

MemGranularity

LRU

LRUTailSize

MeshCache

StagingBufferSize

64K

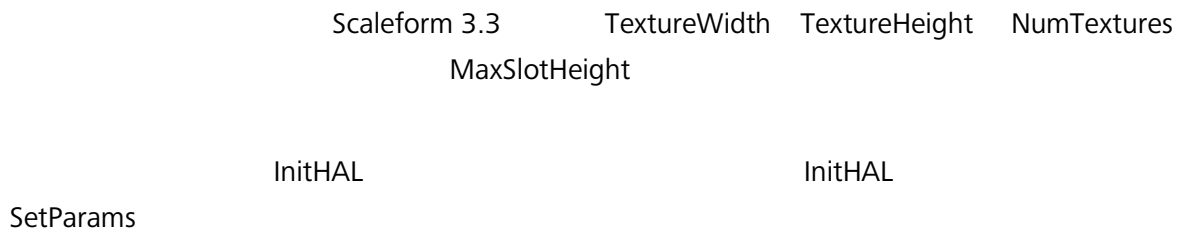
PC

```

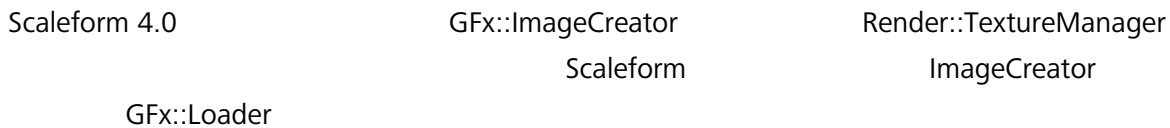
GlyphCacheParams gcp;
gcp.TextureWidth = 1024;
gcp.TextureHeight = 1024;
gcp.NumTextures = 1;
gcp.MaxSlotHeight = 48;

pRenderer->GetGlyphCacheConfig()->SetParams(gcp);

```



2.3

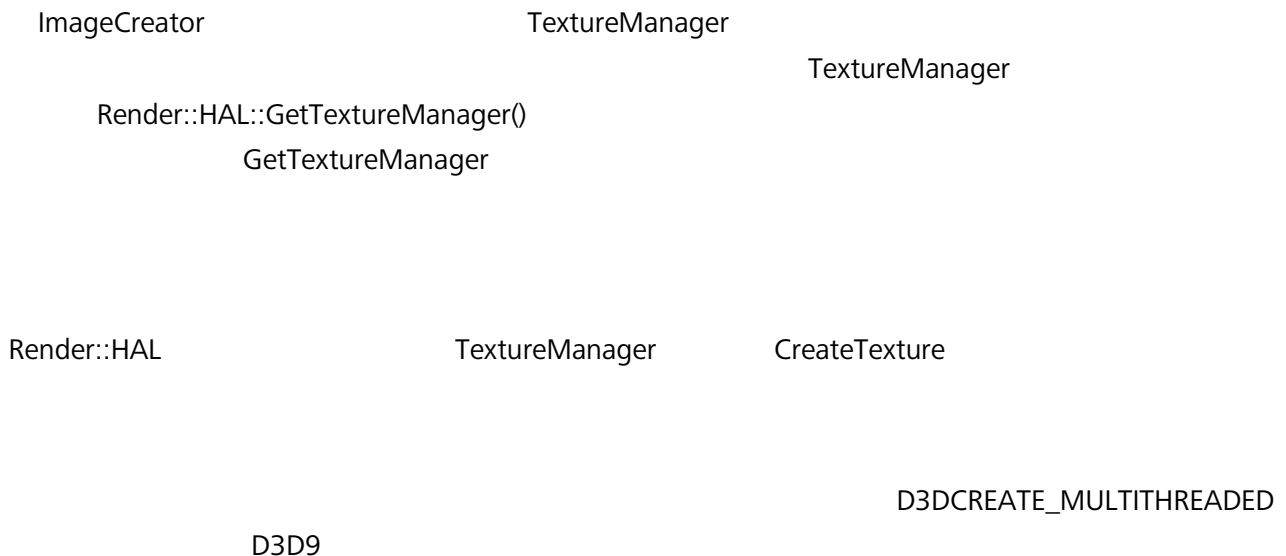


```

Gfx::Loader loader;
...

SF::Ptr<Gfx::ImageCreator> pimageCreator =
    *new Gfx::ImageCreator(pRenderThread->GetTextureManager());
loader.SetImageCreator(pimageCreator);

```

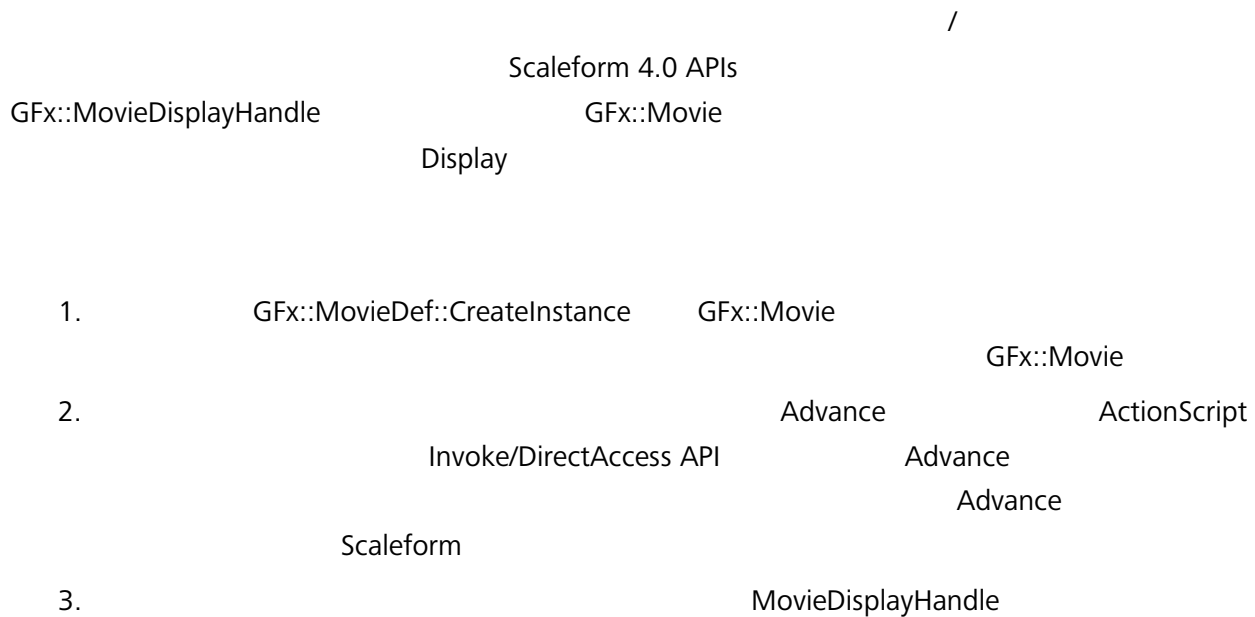


- D3DCREATE_MULTITHREADED D3D9 ThreadCommandQueue Render::HAL

D3D ThreadCommandQueue

Platform::RenderHALThread

2.4



```
//-----
// 1.
Ptr<GfX::Movie>      pMovie = ...;
GfX::MovieDisplayHandle hMovieDisplay;

//
//
pMovie->SetViewport(width, height, 0,0, width, height);
hMovieDisplay = pMovie->GetDisplayHandle();

//                               Scaleform Player
//
pRenderThread->AddDisplayHandle(hMovieDisplay);
```



```

//-----
// 2.

//
//   ExternalInterface      Advance
//   float deltaT = ...;
//   pMovie->HandleEvent(...);
//   pMovie->Advance(deltaT);

//
//
//   pRenderThread->WaitForOutstandingDrawFrame();
//   pRenderThread->DrawFrame();

//-----
// 3.      -      DrawFrame
//   Ptr<Render::Renderer2D> pRenderer = ...;

//   pRenderer->BeginFrame();
//   bool hasData = hMovieDisplay.NextCapture(pRenderer->GetContextNotify());
//   if (hasData)
//       pRenderer->Display(hMovieDisplay);
//   pRenderer->EndFrame();

```

- WaitForOutstandingFrame – CPU
- Advance
- MovieDisplay.NextCapture: “ “Advance false

2.5

Gfx::Movie Release

Render::ThreadCommandQueue Movie
ThreadCommandQueue

Movie::ShutdownRendering(false)
Movie::IsShutdownRenderingComplete
NextCapture false

3

Scaleform

—

GFX::Movie::Advance

ActionScript

Advance

Advance

Scaleform

Advance

Render::HAL

Scaleform

•

•

Scaleform

Player

Scaleform

3.1

Advance

/

Scaleform

GFX::Movie::Advance

GFX::Movie::Capture

NextCapture

- [illegible]

The diagram illustrates the mapping of commands from two different Scaleform versions to a single Movie::Invoke command. On the left, under the heading 'Scaleform 4.0', the command 'Display' is shown. On the right, under the heading 'Scaleform 3.3', the command 'Invoke' is shown. Both 'Display' and 'Invoke' have arrows pointing towards a central 'Movie::Invoke' command. Additionally, there are two 'Advance' commands: one on the far left and one in the middle (between the two main columns), both with arrows pointing towards the central 'Movie::Invoke' command. The 'Capture' command is also shown in the middle column, with an arrow pointing towards the central 'Movie::Invoke' command.

3.2

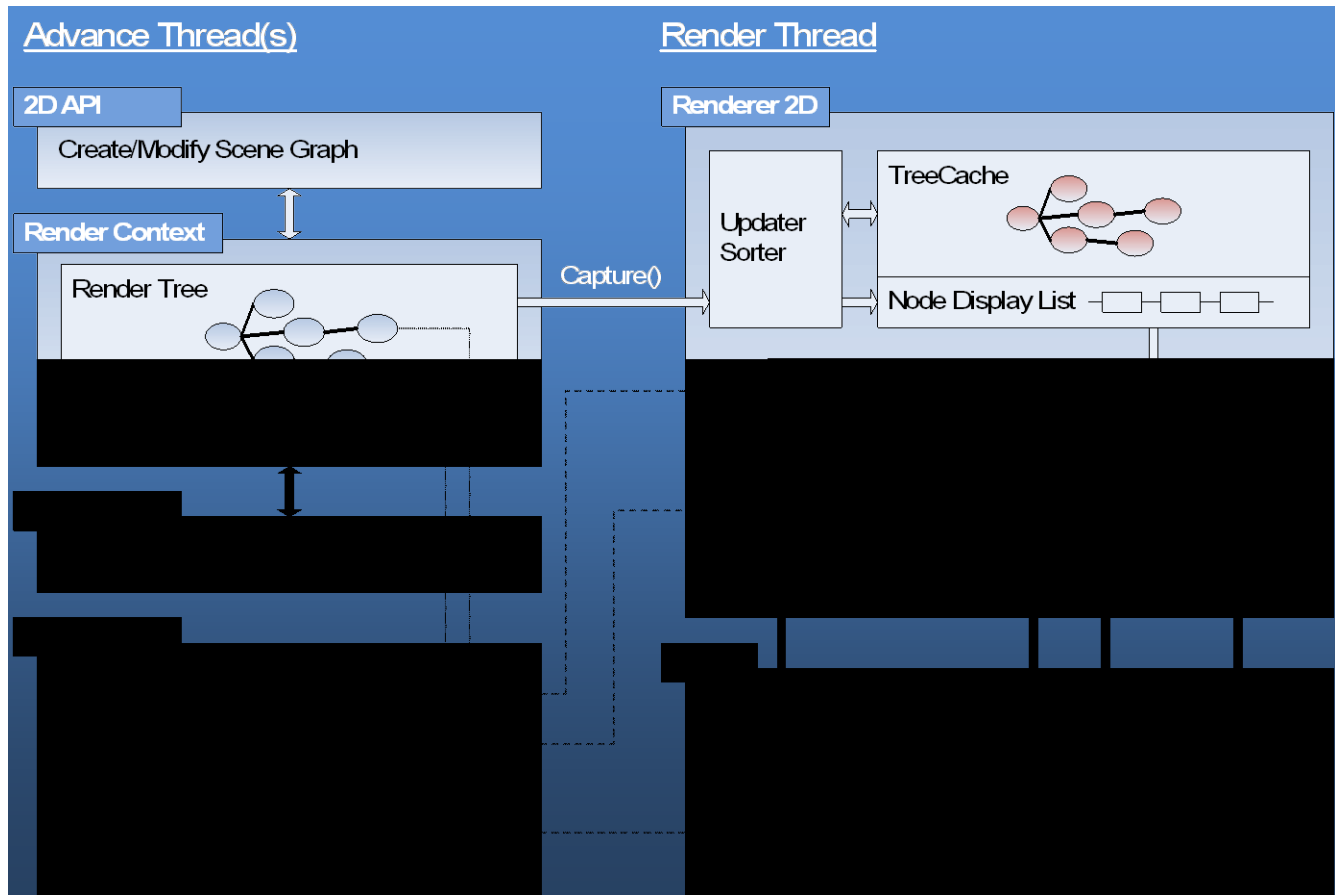
Render::Renderer2D::Display

Render::Renderer2D::Display

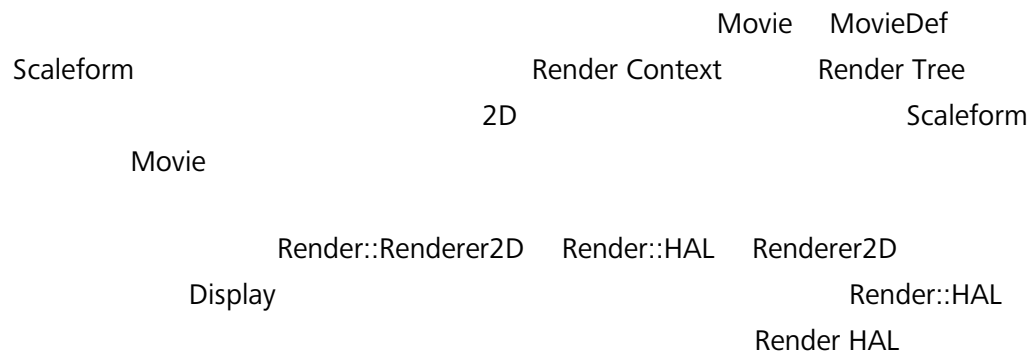
Scaleform

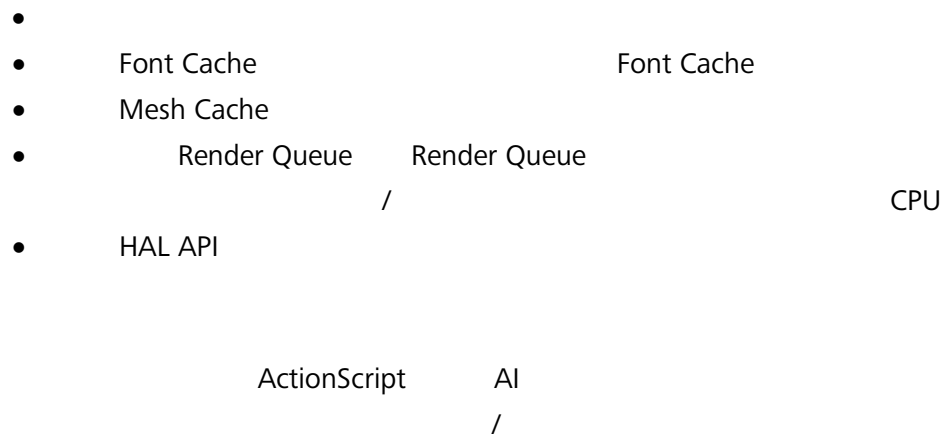
○

Scaleform 4.0



1

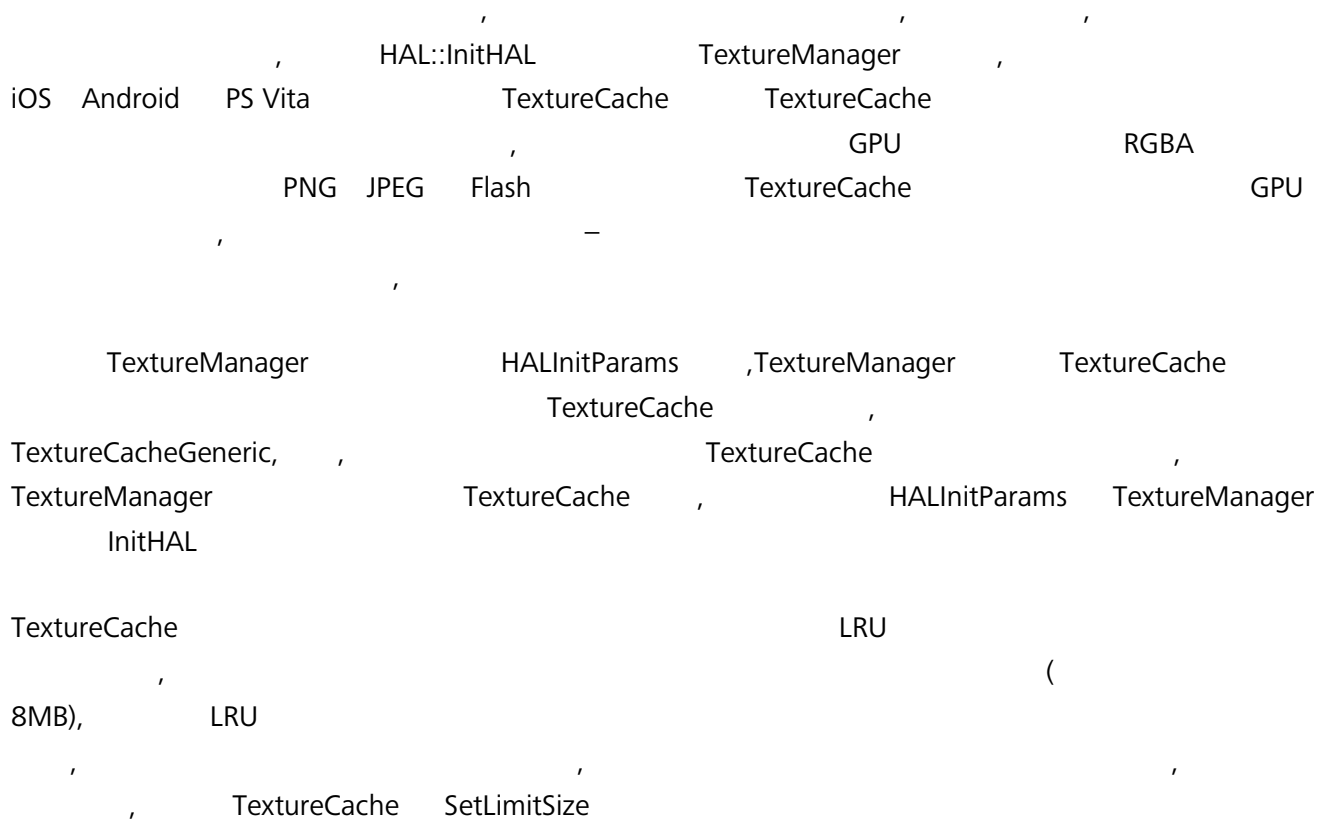




5 渲染子系统说明

5.1 TextureCache

TextureCache



6

Scaleform

6.1

```

    ,
    rendererString      rendererFloat      AS3      setRendererString      AS2
setRendererFloat      AS2/AS3
    ,
    (Float),      HAL::PushUserData      UserDataState::Data      Render::HAL
UserDataStack
    ,
    ,      UserDataState::Data      HAL::PopUserData
    /
    /
```

6.2

```

    ,
    HAL      HAL
    ,
    ,      2D
    ,      AS3
SWF,      m      AS3
```

```
import scaleform.gfx.*;
DisplayObjectEx.setRendererString(m, "Abc");
DisplayObjectEx.setRendererFloat(m, 17.1717);
```

```
4.1
Src/Render/ShaderData.xml ,
```

```
<ShaderFeature id="Position">
  <ShaderFeatureFlavor id="Position2d" hide="true"/>
  <ShaderFeatureFlavor id="Position3d"/>
  <ShaderFeatureFlavor id="Position2dOffset"/>
</ShaderFeature>
```

Position2dOffset

Position2dOffset

```
<ShaderSource id="Position2dOffset" pipeline="Vertex">
```

```
    Position2d(
    attribute float4 pos      : POSITION,
    varying   float4 vpos    : POSITION,
    uniform   float4 mvp[2],
    uniform   float  poffset)
    {
    vpos = float4(0,0,0,1);
    vpos.x = dot(pos, mvp[0]) + poffset;
    vpos.y = dot(pos, mvp[1]) + poffset;
    }
</ShaderSource>
```

Position2d - 'poffset' x y

, D3D9_Shader.cpp , ShaderInterface::SetStaticShader

```
//
    if (pHal->UserDataStack.GetSize() > 0 )
    {
        const UserDataState::Data* data = pHal->UserDataStack.Back();
        if (data->Flags &
(UserDataState::Data::Data_Float|UserDataState::Data::Data_String) &&
            data->RendererString.CompareNoCase("abc") == 0)
        {
//
            ,
            vshader = (VertexShaderDesc::ShaderType)((int)vshader +
VertexShaderDesc::VS_base_Position2dOffset);
            shader = (FragShaderDesc::ShaderType) ((int)shader +
FragShaderDesc::FS_base_Position2dOffset);
        }
    }
}
```

HAL UserDataStack (abc

,) , Position2dOffset

(D3D9_ShaderDescs.h)

ShaderData.xml ,

'poffset'

ShaderInterface::Finish

//


```

if (pHal->UserDataStack.GetSize() > 0 )
{
    const UserDataState::Data* data = pHal->UserDataStack.Back();
    if (data->Flags &
(UserDataState::Data::Data_Float|UserDataState::Data::Data_String) &&
        data->RendererString.CompareNoCase("abc") == 0)
    {
//    poffset
        for ( unsigned m = 0; m < Alg::Max<unsigned>(1, meshCount); ++m)
        {
            float poffset = data->RendererFloat / 256.0f;
            SetUniform(CurShaders, Uniform::SU_poffset, &poffset, 1, 0, m);
        }
    }
}

, 'meshCount'

(
1 HAL , 'm'
),meshCount

```

6.3

```

HAL , Scaleform
,
Scaleform
Scaleform
/ AS2
disableBatching AS3 disableBatching AS2/AS3
,

```

7 GFxShaderMaker

7.1

```
Scaleform 4.1      GFxShaderMaker,      Scaleform
( Render_D3D9) ,
ShaderData.xml
, D3D9 , /
```

Src/Render/D3D9_ShaderDescs.h
Src/Render/D3D9_ShaderDescs.cpp
Src/Render/D3D9_ShaderBinary.cpp

7.2 API HAL

7.2.1 D3D9

D3D9 HAL	ShaderModel 2.0	ShaderModel 3.0	,HAL
----------	-----------------	-----------------	------

GfxShaderMaker '-shadermodel'

GfxShaderMaker.exe -platform D3D1x -shadermodel SM30

ShaderModel 2.0 ShaderModel 2.0

7.2.2 D3D1x

D3D1x HAL	(
D3D11 D3D10.1	
FEATURE_LEVEL_10_0	
FEATURE_LEVEL_9_3	
FEATURE_LEVEL_9_1	

,HAL

GfxShaderMaker '-featurelevel'

GfxShaderMaker.exe -platform D3D1x -featurelevel
FEATURE_LEVEL_10_0,FEATURE_LEVEL_9_1

	FEATURE_LEVEL9_3		FEATURE_LEVEL_9_3
InitHAL,		(	FEATURE_LEVEL_9_1)